Chapter 5. Joint Probability Distributions and Random Sample

Chapter 5: Joint Probability Distributions and Random Sample

- **5.1. Jointly Distributed Random Variables**
- **5.2. Expected Values, Covariance, and Correlation**
- **5. 3. Statistics and Their Distributions**
- **5.4. The Distribution of the Sample Mean**
- **5.5. The Distribution of a Linear Combination**

- The Expected Value of a function h(x,y)

Let X and Y be jointly distribution rv's with **pmf *p*(*x*,*y*)** or **pdf *f*(*x*,*y*)** according to whether the variables are discrete or continuous. Then the expected value of a function ***h*(*X*,*Y*),** **denoted by E[h(X,Y)] or μh(X,Y) ,** is given by

$$E[h(X,Y)] = \begin{cases} \sum_x \sum_y h(x,y) \cdot p(x,y), X \,\&\, Y : discrete \\ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x,y) \cdot f(x,y) dxdy, X \,\&\, Y : continuous \end{cases}$$

- **Example 5.13**

Five friends have purchased tickets to a certain concert. If the tickets are for seats 1-5 in a particular row and the tickets are randomly distributed among the five, what is the expected number of seats separating any particular two of the five?

Solution:

Let X and Y denote the seat numbers of the first and second individuals, respectively. Possible (X,Y) pairs are {(1,2),(1,3),…(5,4),and the joint pmf of (X, Y) is }

$$p(x,y)=\begin{cases}\dfrac{1}{20} & x=1,\ldots,5;\, y=1,\ldots,5;\, x\neq y\\ 0 & \text{otherwise}\end{cases}$$

The number of seats separating the two individuals is

h(X,Y)=|X-Y|-1

Example 5.13 (Cont')

The accompanying table gives h(x,y) for each possible (x,y) pair.

$h(x,y)$		$x=1$	2	3	4	5
y	1	--	0	1	2	3
	2	0	--	0	1	2
	3	1	0	--	0	1
	4	2	1	0	--	0
	5	3	2	1	0	--

$$E[h(X,Y)] = \sum_{(x,y)}\sum h(x,y)\cdot p(x,y)$$

$$= \underset{x \neq y}{\sum_{x=1}^{5}\sum_{y=1}^{5}} (|x-y|-1)\cdot\frac{1}{20} = 1$$

- **Example 5.14**

In Example 5.5, the joint pdf of the amount X of almonds and amount Y of cashews in a 1-lb can of nuts was

$$f(x,y)=\begin{cases}24xy & 0\le x\le 1, 0\le y\le 1, x+y\le 1\\ 0 & \text{otherwise}\end{cases}$$

If 1 lb of almonds costs the company \$1.00, 1 lb of cashews costs \$1.50, and 1 lb of peanuts costs \$0.50, then

(A) The total cost of the contents of a can is ?

(B) The expected total cost is ?

Solution:

(A) $h(X,Y)=(1)X+(1.5)Y+(0.5)(1\text{-}X\text{-}Y)=0.5+0.5X+Y$

(B) The expected total cost is

$$E[h(X,Y)] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h(x,y)\cdot f(x,y)dxdy$$

$$= \int_0^1\int_0^{1-x}(0.5+0.5x+y)\cdot 24xydydx = \$1.10$$

Note: The method of computing $E[h(X_1,\ldots, X_n)]$, the expected value of a function $h(X_1, \ldots, X_n)$ of n random variables is similar to that for two random variables.

5.2 Expected Values, Covariance, and Correlation

- Covariance

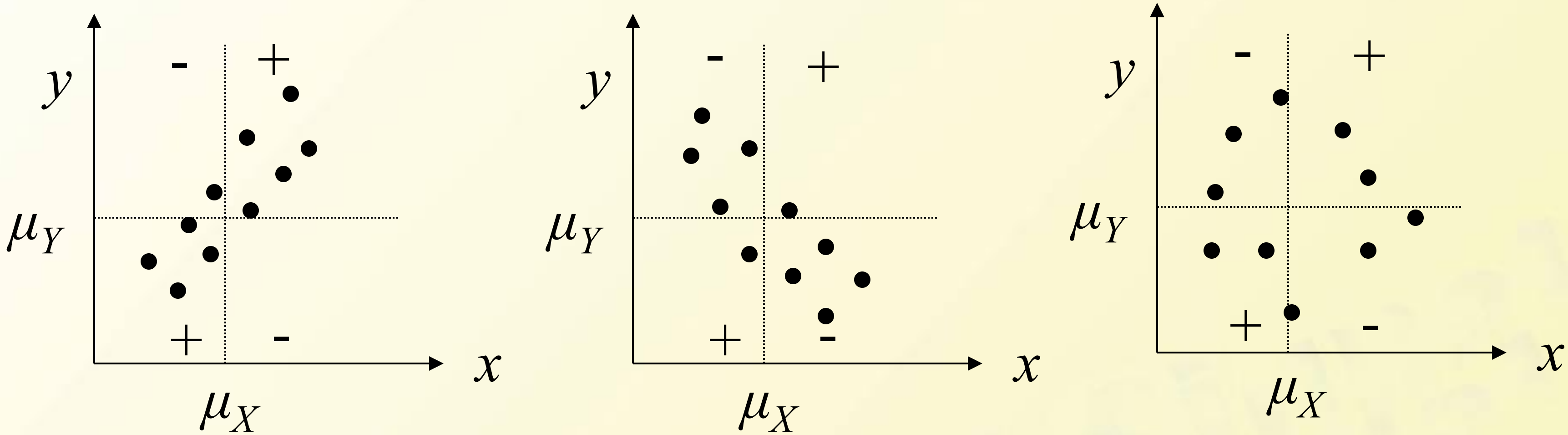
When two rv X and Y are **not independent**, it is frequently of interest to assess **how strongly they are related to one another.**

The Covariance between two rv's X and Y is

$$\mathrm{Cov}(X,Y) = E[(X-\mu_X)(Y-\mu_Y)]$$

$$= \begin{cases} \sum_x \sum_y (x-\mu_X)(y-\mu_Y)\,p(x,y) & X,Y \text{ discrete} \\ \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} (x-\mu_X)(y-\mu_Y)\,f(x,y)\,dx\,dy & X,Y \text{ continuous} \end{cases}$$

- Illustrates the different possibilities.



(a) positive covariance (b) negative covariance; (c) covariance near zero

Here: P(x, y) =1/10

For a strong positive relation ship, Cov(X,Y) should be quite positive.
For a strong negative relation ship, Cov(X,Y) should be quite negative.
If X and Y are not strongly related , Cov(X,Y) near 0

- **Example 5.15**

The joint and marginal pmf's for X = automobile policy deductible amount and Y = homeowner policy deductible amount in Example 5.1 were

$p(x,y)$		y = 0	100	200
x	100	.20	.10	.20
	250	.05	.15	.30

x	100	250
$p_X(x)$	.5	.5

y	0	100	200
$p_Y(y)$	.25	.25	.5

Find cov(X,Y)?

Solution:

From which $\mu_X = \sum x p_X(x) = 175$ and $\mu_Y = 125$.

Therefore

$$\mathrm{Cov}(X,Y) = \sum_{(x,y)}\sum (x-175)(y-125)\,p(x,y)$$

$$= (100-175)(0-125)(0.2) + \ldots + (250-175)(200-125)(0.3)$$

$$= 1875$$

- Proposition

$$\mathrm{Cov}(X, Y) = E(XY) - \mu_X \mu_Y$$

Note: $$\mathrm{Cov}(X, X) = E(X^2) - \mu^2{}_X = V(X)$$

- Correlation

The correlation coefficient of X and Y, denoted by Corr(X,Y), $\rho_{X,Y}$ or just ρ, is defined by

$$\rho_{X,Y} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \cdot \sigma_Y}$$

The normalized version of Cov(X,Y)

Example 5.17

It is easily verified that in the insurance problem of Example 5.15, $\sigma_X = 75$ and $\sigma_Y = 82.92$. This gives

$$\rho = 1875/(75)(82.92)=0.301$$

- Proposition

1. If a and c are either both positive or both negative

$$\text{Corr}(aX+b, cY+d) = \text{Corr}(X,Y)$$

2. For any two rv's X and Y, $-1 \le \text{Corr}(X,Y) \le 1$.

Statement 1 says that the correlation coefficient is not affected by a linear change in the units of measurement

According to Statement 2, the strongest possible positive relationship is evidenced by $\rho=+1$, *whereas* strongest possible negative relationship responds to $\rho=-1$

- Proposition

1. If X and Y are independent, then $\rho = 0$, but $\rho = 0$ does not imply independence.

(when ρ = 0, X and Y are said to be uncorrelated.)

2. $\rho = 1$ or -1 **iff(if and only if)** $Y = aX+b$ for some numbers a and b with $a \neq 0$.

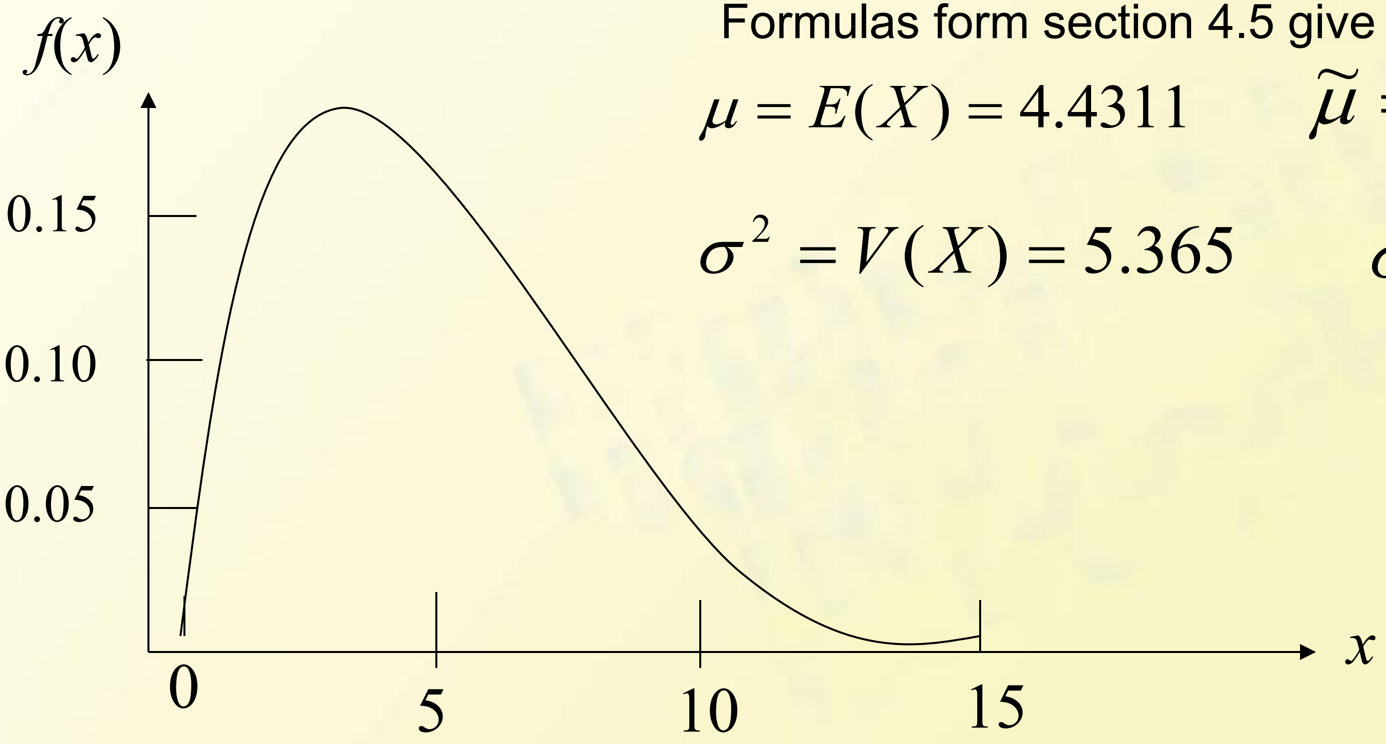
5.3 Statistics and Their Distributions

From this section, we consider function of *n* random variables $X_1, X_2, \ldots, X_n$ focusing especially on their average (X1,X2, …, Xn)/n. we call any such function, itself a random variable, a **statistic**.

As before, we studied some statistics: sample mean, sample standard deviation or sample fourth spread----also varies from sample to sample.

Example 5.19

Given a Weibull Population with α=2, β=5. The corresponding density curve is shown in Fig.5.6.



Formulas form section 4.5 give

$$\mu = E(X) = 4.4311 \qquad \tilde{\mu} = 4.1628$$

$$\sigma^2 = V(X) = 5.365 \qquad \sigma = 2.316$$

5.3 Statistics and Their Distributions

In section 4.5 we studied the Weibull Distribution

- **The Weibull Distribution**

  A random variable X is said to have a Weibull distribution with parameters α and β $(\alpha > 0, \beta > 0)$ if the cdf of X is

$$f(x;\alpha,\beta)=\begin{cases}\dfrac{\alpha}{\beta^{\alpha}}x^{\alpha-1}e^{-(x/\beta)^{\alpha}} & x\geq 0\\ 0 & x<0\end{cases}$$

When α =1, the pdf reduces to the exponential distribution (with λ =1/ β), so the exponential Distribution is a special case of both the gamma and Wellbull distributions.

The Weibull Distribution

- Mean and Variance

$$\mu = \beta\Gamma\left(1+\frac{1}{\alpha}\right); \quad \sigma^2 = \beta^2\left\{\Gamma\left(1+\frac{2}{\alpha}\right)-\left[\Gamma\left(1+\frac{1}{\alpha}\right)\right]^2\right\}$$

- The cdf of a Weibull Distribution

$$F(x;\alpha,\beta)=\begin{cases} 0 & x<0 \\ 1-e^{-(x/\beta)^{\alpha}} & x\geq 0 \end{cases}$$

5.3 Statistics and Their Distributions

- ## Example 5.19 (Cont')

We used MINITAB to generate six different samples, each with *n=10*.

Sample	1	2	3	4	5	6
1	6.1171	5.07611	3.46710	1.55601	3.12372	8.93795
2	4.1600	6.79279	2.71938	4.56941	6.09685	3.92487
3	3.1950	4.43259	5.88129	4.79870	3.41181	8.76202
4	0.6694	8.55752	5.14915	2.49795	1.65409	7.05569
5	1.8552	6.82487	4.99635	2.33267	2.29512	2.30932
6	5.2316	7.39958	5.86887	4.01295	2.12583	5.94195
7	2.7609	2.14755	6.05918	9.08845	3.20938	6.74166
8	10.2185	8.50628	1.80119	3.25728	3.23209	1.75486
9	5.2438	5.49510	4.21994	3.70132	6.84426	4.91827
10	4.5590	4.04525	2.12934	5.50134	4.20694	7.26081

Example 5.19 (Cont')

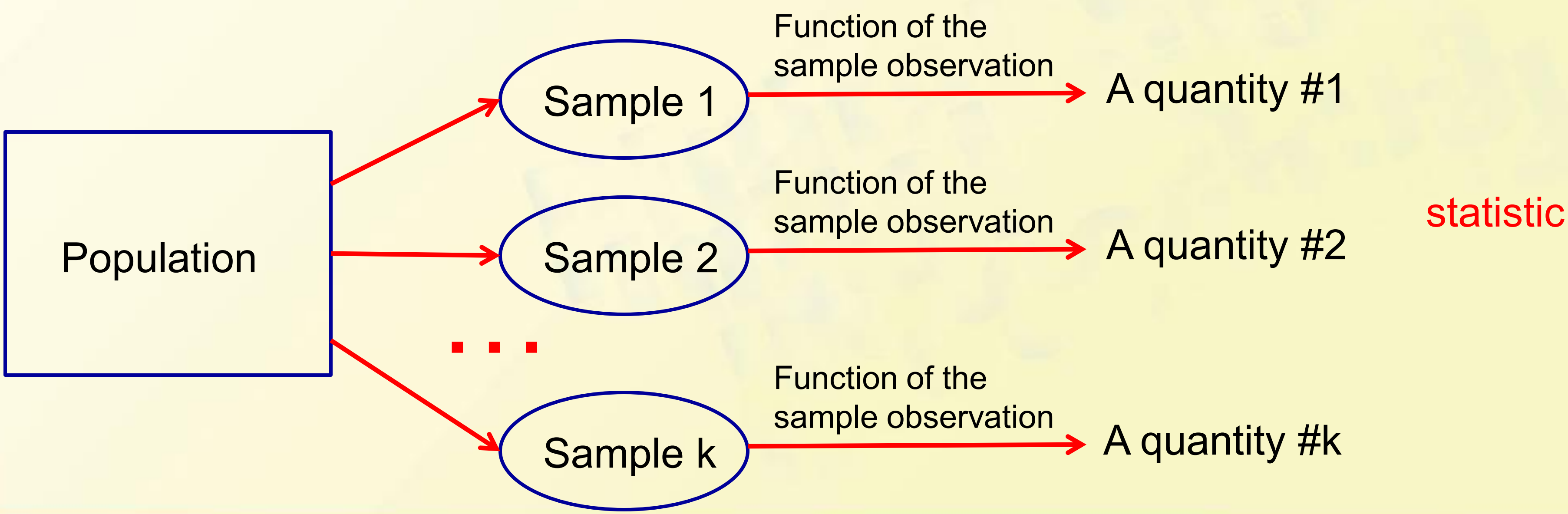
Sample	1	2	3	4	5	6
Mean	4.401	5.928	4.229	4.132	3.620	5.761
Median	4.360	6.144	4.608	3.857	3.221	6.342
Standard Deviation	2.642	2.062	1.611	2.124	1.678	2.496

For sample mean, none of the estimates from these six samples is identical to what is being estimated($\mu = 4.4311$) .

The estimates from the second and sixth samples are much too large, whereas the fifth sample gives a substantial underestimate. All six of the resulting estimates are in error by at least a small amount

5.3 Statistics and Their Distributions

In summary, the value of the individual sample observations vary form sample to sample, so in general the value of any quantity computer from sample data, and the value of a sample characteristic used as an estimate of the corresponding population characteristic, will virtually never coincide with what is being estimated.



5.3 Statistics and Their Distributions

- **Statistic**

  A statistic is any quantity whose value can be calculated from **sample data** (with a function).

- Prior to obtaining data, there is **uncertainty** as to what value of any particular statistic will result. Therefore, **a statistic is a random variable.** A **statistic** will be denoted by an uppercase letter; a lowercase letter is used to represent the calculated or observed value of the statistic.

- The probability distribution of a statistic is sometimes referred to as its **sampling distribution**. It describes how the statistic varies in value across all samples that might be selected.

5.3 Statistics and Their Distributions

- The probability distribution of any particular statistic depends on

1. The population distribution, *e.g.* the normal, uniform, etc. , and the corresponding parameters
2. The sample size n (refer to Ex. 5.20 & 5.30)
3. The method of sampling, *e.g.* sampling with **replacement** or **without replacement**

- **Example**

  Consider selecting a sample of size $n = 2$ from a population consisting of just the three values 1, 5, and 10, and suppose that the statistic of interest is the sample variance.

➢ If sampling is done "**with replacement**", then $S^2 = 0$ will result if $X_1 = X_2$.

➢ If sampling is done "**without replacement**", then S^2 can not equal 0.

5.3 Statistics and Their Distributions

- Random Sample

The rv's $X_1, X_2, \ldots, X_n$ are said to form **a (simple) random sample** of size n if

1. The Xi's are independent rv's.
2. Every Xi has the same probability distribution.

When conditions 1 and 2 are satisfied, we say that the X_i's are independent and identically distributed (**i.i.d**)

Note: Random sample is one of commonly used sampling methods in practice.

5.3 Statistics and Their Distributions

- Random Sample

➢ Sampling with **replacement** or from an **infinite population** is random sampling

➢ Sampling **without replacement** from a finite population is generally considered **not random sampling**. However, if the sample **size *n* is much smaller than the population size *N* ($n/N \le 0.05$), it is approximately random sampling.**

Note: The virtue of random sampling method is that the probability distribution of any statistic can be more easily obtained than for any other sampling method.

5.3 Statistics and Their Distributions

- Deriving the Sampling Distribution of a Statistic

> Method #1: Calculations based on probability rules
>
> ***e.g.*** Example 5.20 & 5.21

> Method #2:
>
> Carrying out a simulation experiments
>
> ***e.g.*** Example 5.22 & 5.23

Example 5.20

A large automobile service center charges \$40, \$45, and \$50 for a tune-up of four-, six-, and eight-cylinder cars, respectively. If 20% of its tune-ups are done on four-cylinder cars, 30% on six-cylinder cars, and 50% on eight-cylinder cars, then the probability distribution of revenue from a single randomly selected tune-up is given by

x	40	45	50
$p(x)$	0.2	0.3	0.5

$\mu = 46.5$

$\sigma^2 = 15.25$

Suppose on a particular day only two servicing jobs involve tune-ups.

Let X_1 = the revenue from the first tune-up &

X_2 = the revenue from the second,

which constitutes a random sample with the above probability distribution.

5.3 Statistics and Their Distributions

- **Example 5.20 (Cont')**

x_1	x_2	$p(x_1,x_2)$	$\bar{x}$	s^2
40	40	0.04	40	0
40	45	0.06	42.5	12.5
40	50	0.10	45	50
45	40	0.06	42.5	12.5
45	45	0.09	45	0
45	50	0.15	47.5	12.5
50	40	0.10	45	50
50	45	0.15	47.5	12.5
50	50	0.25	50	0

Known the Population Distribution

x	40	42.5	45	47.5	50
$p_x(x)$	0.04	0.12	0.29	0.30	0.25

$$\mu_{\overline{X}} = E(\overline{X}) = 46.5 = \mu$$

$$\sigma_{\overline{X}}^2 = V(\overline{X}) = \sum \bar{x}^2 \cdot P_{\overline{X}}(\bar{x}) - \mu_{\overline{X}}^2 = 7.625 = \frac{15.25}{2} = \frac{\sigma^2}{2}$$

The variance of $\overline{X}$ is precisely half that of the original variance (because n=2)

Similarly,

$$P_{S^2}(50) = P(S^2 = 50) = P(X_1 = 40, X_2 = 50 \ \ or \ \ X_1 = 50, X_2 = 40) = 0.10 + 0.10 = 0.20$$

s^2	0	12.5	50
$p_{s^2}(s^2)$	0.38	0.42	0.20

$$\mu_{S^2} = E(S^2) = \sum s^2 \cdot P_{\overline{X}}(\overline{x}) - \mu_{\overline{x}}^2 = (0)(0.38) + (12.5)(0.42) + (50)(0.20) = 15.25 = \sigma^2$$

That is, the $\overline{X}$ Sampling distribution is centered at the population mean μ

And the S^2 Sampling distribution is centered at the population variance σ^2

5.3 Statistics and Their Distributions

- **Example 5.20 (Cont')**

$\bar{x}$	40	42.5	45	47.5	50
$p_{\bar{x}}(x)$	0.04	0.12	0.29	0.30	0.25

n=2

$\bar{x}$	40	41.25	42.5	43.75	45	43.26	47.5	48.75	50
$p_{\bar{x}}(x)$	0.0016	0.0096	0.0376	0.0936	0.1761	0.2340	0.2350	0.1500	0.0625

n=4

...

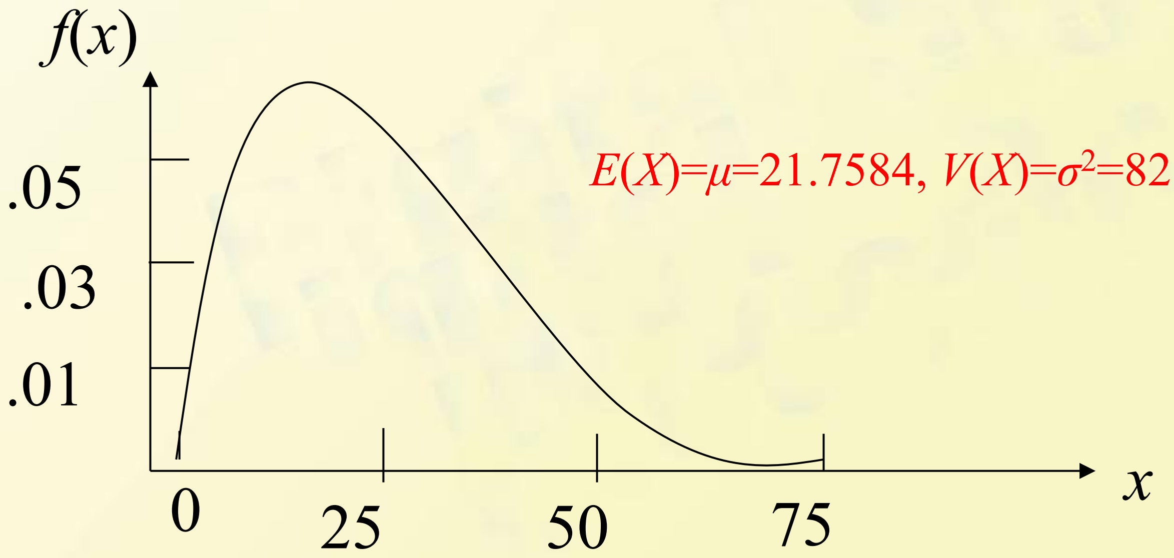
Simulation Experiments

This method is usually used when a derivation via probability rules is too difficult or complicated to be carried out. Such an experiment is virtually always done with the aid of a computer. And the following characteristics of an experiment must be specified:

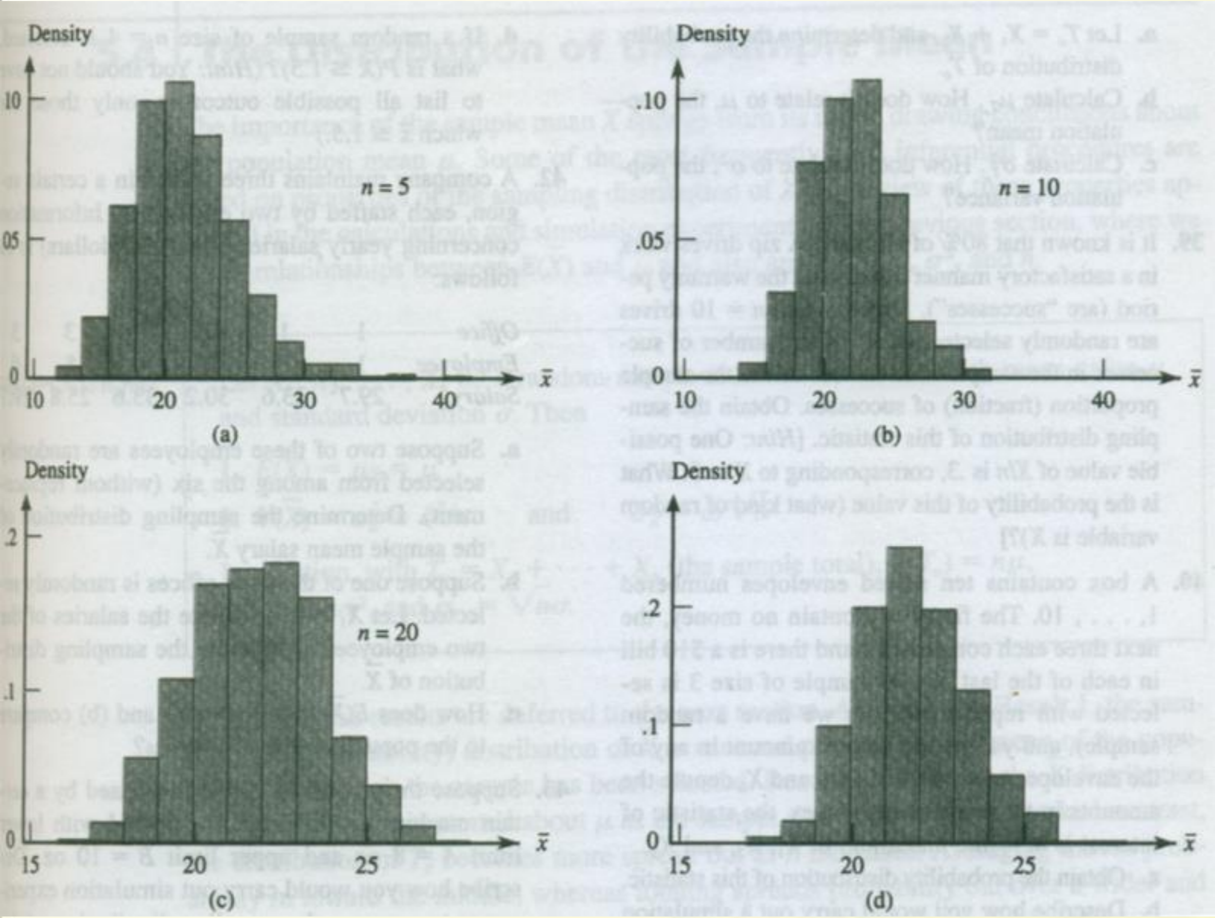
- The statistic of interest (*e.g.* sample mean, S, etc.)
- The population distribution (normal with $\mu = 100$ and $\sigma = 15$, uniform with lower limit $A = 5$ and upper limit $B = 10$, etc.)
- The sample size n (*e.g.*, $n = 10$ or $n = 50$)
- The number of replications k (*e.g.*, $k = 500$ or 1000) (the actual sampling distribution emerges as $k \to \infty$)

Example 5.23

Consider a simulation experiment in which the population distribution is quite skewed. Figure shows the density curve of a certain type of electronic control (actually a **lognormal distribution with $E(\ln(X)) = 3$ and $V(\ln(X))=.4$**).



Example 5.23 (Cont')



1. **Center of the sampling distribution remains at the population mean.**
2. **As *n* increases:**
   - ✓ **Less skewed ("more normal")**
   - ✓ **More concentrated ("smaller variance")**

Sample histogram for $\overline{X}$ based on 500 samples ,each consisting of n observations:

(a) n=5; (b) n=10; (C) n=20; (d) n=30